

1

DATA PREPARATION & BASELINE

2,000 training questions (TriviaQA) + **200 held-out** for evaluation. Base model evaluated at Gen 0 to establish **K₀** factual baseline per backbone.

Qwen 2.5 1.5B (K₀=78)

Gemma 3 1B (K₀=46)

Gemma 4 E2B (K₀=76)

2

RECURSIVE TRAINING (REPLACE-ONLY)

Each generation: fresh adapter trained on previous generation's synthetic outputs. **10 recursive generations**, strict data replacement (no accumulation).

QLoRA (4-bit NF4)

Full Fine-Tuning

3-5 seeds

3

PRESSURE MANIPULATION (5 AXES)

Systematic variation of effective training pressure along independent experimental axes.